
Credit Risk Prediction — Project Documentation

Task 2 — DevelopersHub Data Science Internship
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Objective

Predict whether a loan applicant is likely to get approved or rejected based on personal and financial information.

Dataset

- Source: Kaggle — Loan Prediction Dataset
- Shape: 614 rows × 13 columns
- Target: Loan_Status (Y = Approved, N = Rejected)
- Class Distribution: 422 Approved (68.7%), 192 Rejected (31.3%) — imbalanced

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Step 1 — Missing Value Analysis

Column	Missing
Credit_History	50
Self_Employed	32
Dependents	15
Loan_Amount_Term	14
LoanAmount	22
Gender	13
Married	3

Step 2 — Data Cleaning

- Numerical columns (LoanAmount, Loan_Amount_Term, Credit_History) → filled with **median**
- Categorical columns (Gender, Married, Dependents, Self_Employed) → filled with **mode**
- Result: Zero missing values ✅

Step 3 — EDA Key Findings

- Most loans are between 100-200 (thousands) — right skewed distribution
- Income alone doesn't determine approval — both approved and rejected applicants have similar income ranges
- Graduates get approved significantly more than non-graduates
- Dataset is imbalanced — model may be biased toward approvals

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Step 4 — Feature Engineering

- Dropped Loan_ID (irrelevant)
- Label Encoded all categorical columns: Gender, Married, Dependents, Education, Self_Employed, Property_Area, Loan_Status
- Applied StandardScaler to normalize all features — needed because income values (up to 81,000) and credit history (0 or 1) are on completely different scales

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Step 5 — Model Training

- Algorithm: **Logistic Regression**
- Split: 80% train (491 samples), 20% test (123 samples)
- Applied StandardScaler before training to fix convergence issue

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Step 6 — Model Evaluation

- ✅ **Accuracy: 78.86%**

- Approved recall: 0.99 — almost never misses an approval
- Rejected recall: 0.42 — struggles to identify rejections
- **25 false positives** — approved applicants who should have been rejected (key business risk)
- **1 false negative** — rejected applicant who should have been approved

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Conclusion

The model performs well overall at 78.86% accuracy but struggles with identifying rejections due to class imbalance. In a real banking scenario, the 25 false positives represent actual financial risk — applicants approved who may default. Future improvement would involve balancing the dataset using SMOTE and testing Random Forest for better rejection detection.

Status: TASK 2 ✅ COMPLETE